1/26/2022

Wednesday, January 26, 2022 11:25 AM

M168_Notes_2.3

2.3 Systems of Linear Equations: Underdetermined and Overdetermined Systems

Example:

$$\begin{aligned} x + 2y + z &= -2 \\ -2x - 3y - z &= 1 \\ 2x + 4y + 2z &= -4 \end{aligned}$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & -2 \\ -2 & -3 & -1 & 1 \\ 2 & 4 & 2 & -4 \end{array}\right] \xrightarrow{\text{RREF}} \begin{array}{c} \begin{array}{ccc} x & y & z \end{array} \\ \left[\begin{array}{ccc|c} 1 & 0 & -1 & 4 \\ 0 & 1 & 1 & -3 \\ 0 & 0 & 0 & 0 \end{array}\right] \end{array}$$

$$\begin{aligned} x - t &= 4 \\ y + t &= -3 \\ 0 &= 0 \end{aligned}$$

$$\begin{aligned} x &= 4 + t \\ y &= -3 - t \\ z &= t \end{aligned}$$

$(4+t, -3-t, t), \forall t \in \mathbb{R}$

Now check is this in RREF?

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 4 \\ 0 & 1 & 1 & -3 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Are the rows of all zeros is below the non-zero rows?

Is the first non-zero entry in any row is a 1?

Do the leading 1's go down in a diagonal?

If a column has a leading 1 then does the rest of the column have zeros?

A column containing a leading 1 is called a *unit column* and the variable associated with the column is a *basic variable.*

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 4 \\ 0 & 1 & 1 & -3 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Example:

$$\begin{aligned} x + y - 2z &= -3 \\ 2x - y + 3z &= 7 \\ x - 2y + 5z &= 0 \end{aligned} \longrightarrow \left[\begin{array}{ccc|c} 1 & 1 & -2 & -3 \\ 2 & -1 & 3 & 7 \\ 1 & -2 & 5 & 0 \end{array}\right] \xrightarrow{\text{RREF}} \left[\begin{array}{ccc|c} 1 & 0 & 0.33 & 0 \\ 0 & 1 & -2.33 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

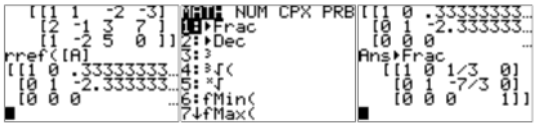
x y z

$1/3$

$-7/3$

$$\boxed{0 = 1}$$

No solution.



RREF form?

Are the rows of all zeros below the non-zero rows? yes
Is the first non-zero entry in any row is a 1? yes
Do the leading 1's go down in a diagonal? yes
If a column has a leading 1 then is the rest of the column is zero? yes.

Number of Solutions Theorem

Case 1: If the number of equations is greater than or equal to the number of variables in a linear system then one of the following is true:

- The system has no solution
- The system has exactly one solution
- The system has infinitely many solutions

Case 2: If there are fewer equations than variables then the system has no solution or infinitely many solutions.

Example: Solve the following system (Case 2)

$$\begin{aligned} x_1 + 2x_2 + 4x_3 &= 2 \\ x_1 + x_2 + 2x_3 &= 1 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 1 \end{array}\right] \rightsquigarrow \begin{array}{c} \begin{array}{ccc} x_1 & x_2 & x_3 \end{array} \\ \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 \end{array}\right] \end{array}$$

$x_1 = 0$

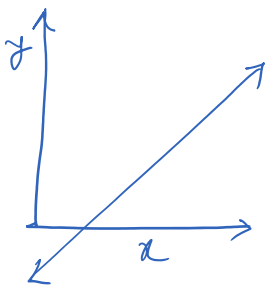
$x_2 + 2x_3 = 1$

$x_3 = t \in \mathbb{R}$

$x_2 = 1 - 2t$

$\boxed{(0, 1-2t, t), \text{ For all } t \in \mathbb{R}}$

$(0, r, \frac{1-r}{2})$



Example: Solve the following system (Case 1):

$$\begin{aligned} 4x + 6y &= 8 \\ 3x - 2y &= -7 \end{aligned}$$

$$\begin{aligned} x + 3y &= 5 \\ 2x + 6y &= 10 \end{aligned}$$

$$\left[\begin{array}{cc|c} 4 & 6 & 8 \\ 3 & -2 & -7 \end{array}\right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & -1 \\ 0 & 1 & 2 \end{array}\right] \rightarrow (-1, 2)$$

2x3 Error

$$\left[\begin{array}{cc|c} 1 & 3 & 5 \\ 0 & 0 & 0 \end{array}\right]$$

$x + 3y = 5$, $y = t$,

$x = 5 - 3y$

$(5-3t, t) = \boxed{(-1, 2)}$

$x = 5 - 3t$

$y = t$

$t = 2$

$5 - 6 = -1$

$(5-4t, t)$ $(-1, 2)$

$5 - 4 \cdot 2 = -3 \neq -1$

A company is buying three kinds of vehicles. Carts hold 3 people and cost \$9,000, vans hold 8 people can cost \$27,000 and minivans hold 7 people and cost \$27,000. The company needs to seat 48 people and has \$162,000 to purchase vehicles. How many of each type of vehicle can be purchased?